

TrES-2b

R-1019

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_160504 · created 2026-07-30T04:26:47+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457512.84742 +/- 0.00044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.093 +/- 0.015
Transit depth (Rp/Rs)^2	0.87 +/- 0.28 [%]
Orbital Inclination (inc)	88.6 +/- 1.45
Airmass coefficient 1 (a1)	1.0174 +/- 0.01
Airmass coefficient 2 (a2)	-0.0106 +/- 0.003
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	#2 - [109, 282]
Optimal Aperture	3.84
Optimal Annulus	8.36
Transit Duration (day)	0.1055 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.093 $\pm$ 0.015 vs 0.1254 (deviation 1.93 σ)
Duration fit vs published	0.1055 d vs 0.0746 d (deviation 5.22 σ)
Mid-time O-C	-1.1 min
Depth SNR	5.6
Residual scatter	0.98 %

Light curve

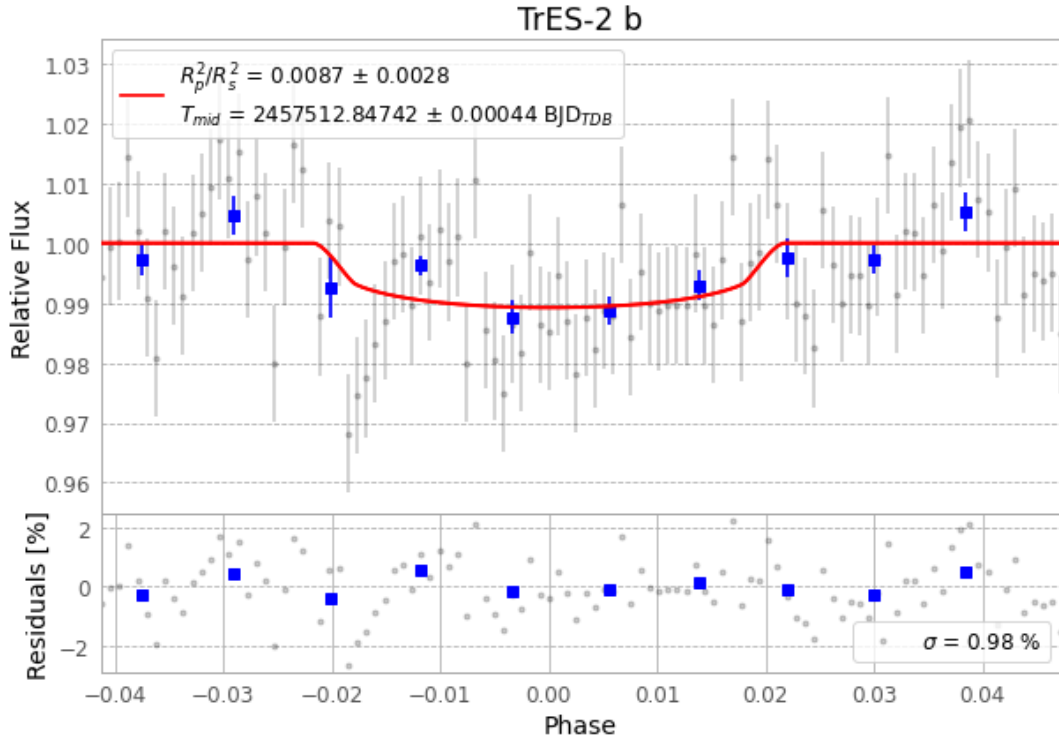


Figure 1 — FinalLightCurve_TrES-2 b_2016-05-03.png (SHA-256 a0cc43ebca69a8b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c9b224b789443640...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

106 FITS frames (70 MB), TRES-2160504055112.FITS → TRES-2160504110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-160504.log (c1dd1e712cc0...), FinalLightCurve_TrES-2 b_2016-05-03.png (a0cc43ebca69...), FinalLightCurve_TrES-2 b_2016-05-03.csv (bf625ac5e251...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 04:26Z.